

Modeling the regulation of the competence-evoking quorum sensing network in *Streptococcus pneumoniae*

Diana Karlsson^{a,c,*}, Stefan Karlsson^b, Erik Gustafsson^{a,d},
Birgitta Henriques Normark^c, Patric Nilsson^a

^a Molecular Biology, School of Life Sciences, University of Skövde, SE-541 28 Skövde, Sweden

^b Mathematics, School of Life Sciences, University of Skövde, SE-541 28 Skövde, Sweden

^c Swedish Institute for Infectious Disease Control, Department of Microbiology, Tumor and Cell Biology, Karolinska Institutet, SE-171 77 Stockholm, Sweden

^d Department of Microbiology, Tumor and Cell Biology, Karolinska Institutet, SE-171 77 Stockholm, Sweden

Received 10 May 2006; received in revised form 15 August 2006; accepted 16 August 2006

Abstract

Competence for genetic transformation seems to play a fundamental role in the biology of *Streptococcus pneumoniae* and is believed to account for serotype switching, evolution of virulence factors, and rapid emergence of antibiotic resistance. The initiation of competence is regulated by the quorum sensing system referred as the ComABCDE pathway. Experimental studies reveal that competence is down-regulated a short time after its induction and several hypotheses about the mechanism(s) responsible for this shut-down have been presented. Possibly, a ComX-dependent gene product, such as a repressor or a phosphatase, is involved. To better understand the down-regulation of the competence-evoking system in *S. pneumoniae*, a mathematical model was set up. By analyzing the model, we suggest that shut-down of competence possibly occurs at the transcriptional level on the *comCDE* operon. As a result of introducing a putative *comX*-dependent repressor, which inhibits expression of *comCDE* and *comX*, in the mathematical model, competence is demonstrated to appear in waves. This is supported by experimental studies showing the appearance of successive competence cycles in pneumococcal batch cultures.

© 2006 Elsevier Ireland Ltd. All rights reserved.

Keywords: Quorum sensing; Competence; Mathematical modeling; *Streptococcus pneumoniae*; Hysteresis; Oscillations

1. Introduction

Streptococcus pneumoniae is the leading cause of community-acquired infections worldwide and is responsible for sicknesses ranging from mild upper respiratory infections to severe invasive diseases like meningitis and bacteremia. The global success of this

bacterium might in part be explained by its genetic plasticity, which is mediated at least partly by its quorum sensing controlled competence for genetic transformation, i.e., the physiological state in which exogenous DNA can be internalized and incorporated in the bacterial chromosome. By providing each pneumococcal cell with access to DNA released from other cells, transformation allows competent cells to acquire new traits. Possibly, competence has influenced both the evolution of virulence factors (Coffey et al., 1991, 1998; Hollingshead et al., 2000; Kelly et al., 1994; Ween et al., 1999) and the rapid emergence of antibiotic resistance

* Corresponding author at: Molecular Biology, School of Life Sciences, University of Skövde, SE-541 28 Skövde, Sweden.
Tel.: +46 500 448635; fax: +46 500 448499.

E-mail address: diana.karlsson@his.se (D. Karlsson).

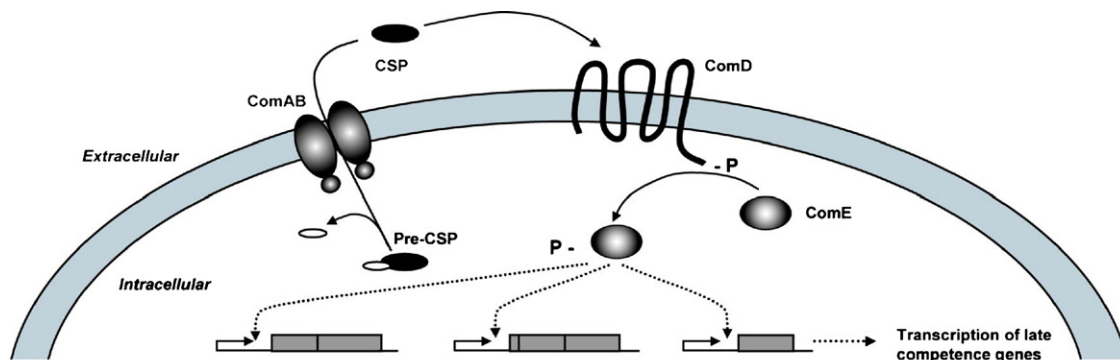


Fig. 1. Illustration of the quorum sensing regulated ComABCDE pathway in *S. pneumoniae*. The quorum sensing signal is a heptadecapeptide called CSP (competence-stimulating peptide) which is derived from its precursor ComC through cleavage and export by an ABC-transporter, ComAB. CSP accumulates extracellularly and is sensed by a histidine kinase, ComD, which autophosphorylates upon binding CSP. ComD then passes the signal further by transferring the phosphate group to its cognate response regulator, ComE, which acts as a transcriptional activator and enhances the expression of *comAB*, *comCDE*, and *comX*. Transcription of a large set of competence-specific genes depends in turn on ComX, an alternative sigma factor. In the figure, flows of the components are indicated by solid lines, whereas dashed lines denote influences of the components.

in *S. pneumoniae* (Coffey et al., 1991; Dowson et al., 1989; Hakenbeck et al., 1998). The transforming DNA probably originates from the lysis of non-competent pneumococcal cells triggered by competent cells (Guiral et al., 2005). It has been shown that this DNA release continues for several hours after the disappearance of competence (Moscoso and Claverys, 2004), indicating that this cell lysis might serve other purposes than genetic exchange. Guiral et al. (2005) proposed that competence thereby may also reinforce virulence by coordinating the release of virulence factors. This is supported by studies demonstrating attenuated virulence in competent-deficient mutants (Bartilson et al., 2001; Lau et al., 2001).

Analogous to a wide range of bacterial adaptive responses, such as sporulation, virulence, and bioluminescence, the initiation of competence in streptococci belonging to the mitis and anginosus phylogenetic groups is controlled by a so called quorum sensing system (Dunny and Leonard, 1997). Gram-negative as well as gram-positive bacteria employ this sort of chemical communication system that helps monitoring cell density and regulates group behavior. The peptide-controlled quorum sensing system initiating competence in pneumococci is referred to as the ComABCDE pathway (Fig. 1). The unmodified competence-stimulating peptide (CSP) is encoded by the 3' moiety of *comC* (Håvarstein et al., 1995) and exported by the dedicated ABC transporter ComAB (Hui et al., 1995), which recognizes and processes the amino-terminus of ComC. CSP accumulates extracellularly and is detected by the membrane-bound histidine kinase receptor, ComD (Pestova et al., 1996). The binding of CSP is suggested

to cause a conformational change in ComD, leading to autophosphorylation of the histidine kinase. ComD then probably transfers the phosphate group to its cognate response regulator, ComE, which acts as a transcriptional activator recognizing specific DNA-binding sites of its target promoters (Ween et al., 1999). The phosphorylation event enhances the affinity of ComE to its binding sites resulting in the increased expression of the target genes (i.e., early *com* genes). ComE-binding sites have been identified upstream of both *comCDE* and *comAB* operons (Ween et al., 1999), and of *comX*, which encodes an alternative sigma factor. ComX has been found to be a unique link between quorum sensing and competence development (Luo et al., 2003); expression of a large set of competence-specific operons depends on ComX (Peterson et al., 2004), which recognizes a consensus promoter sequence termed *cin*-box when associated with the RNA polymerase holoenzyme (Campbell et al., 1998; Lee and Morrison, 1999; Luo and Morrison, 2003). These late *com* genes encode proteins involved in binding, uptake, processing, and recombination of exogenous DNA (Campbell et al., 1998; Martin et al., 1995; Pestova and Morrison, 1998).

In pneumococci, competence is induced when CSP reaches an external concentration of $1\text{--}10\text{ ng ml}^{-1}$ (Håvarstein et al., 1995) corresponding to a cell density varying from 10^5 to $5 \times 10^8\text{ cells ml}^{-1}$ (Chen and Morrison, 1998). CSP-induced competence affects almost 100% of the non-competent cells in an exponential phase of growth (Håvarstein et al., 1995). Waves of competence may occur in a culture before reaching a stationary phase. In the second wave, only about 12% of the cells become competent and any additional waves

Table 1
Description of the variables

Variable	Description	Unit	Localization
X_{ComC}	Total amount of precursor CSP	nmol	Intracellular
X_{ComD}	Total amount of non-complexed histidine kinase	nmol	Membrane bound
X_{CSP}	Concentration of CSP	pg ml ⁻¹	Extracellular
X_{ComDP}	Total amount of complexed histidine kinase	nmol	Membrane bound
X_{ComE}	Total amount of response regulator	nmol	Intracellular
$X_{\text{ComE~P}}$	Total amount of phosphorylated response regulator	nmol	Intracellular
X_{ComX}	Total amount of alternative sigma factor	nmol	Intracellular
X_{R}	Total amount of putative repressor	nmol	Intracellular

involve only a minority of the cells (Chen and Morrison, 1998; Morrison, 1997). Over recent years, the understanding of the initiation of competence in pneumococci at the molecular level has made considerable progress, although the mechanism(s) responsible for the abrupt shut-off of competence a short time after its induction is still unknown and remains to be elucidated.

The primary aim of this paper was to point out a possible shut-off mechanism of competence. Since experimental studies have not yet succeeded in identifying a negative control mechanism, a theoretical approach seemed suitable for suggesting such a mechanism of this quorum sensing controlled network. Therefore, a mathematical model of the competence-evoking quorum sensing system in *S. pneumoniae* was set up. Our *in silico* data show that the down-regulation of competence possibly occur at transcriptional level by a putative repressor that inhibits expression of *comCDE* and *comX*. Furthermore, including such a repressor in the system, the mathematical model envisaged waves of competence as observed in experimental studies.

2. Results

2.1. Initial model

In this paper, a dynamical system of non-linear ordinary differential equations (ODEs) describing the components involved in the competence-evoking quorum sensing pathway is set up. A number of assumptions are made in order to simplify our model. First, the *in silico* bacterial population modeled herein is assumed being homogenous. Second, the mRNA level is approximated to be in quasi-steady state as in Dockery and Keener (2001). This standard quasi-steady state assumption relies upon the fact that mRNA turnover is very fast in comparison to protein turnover. The half-life of mRNA in bacteria is often less than a minute at 37 °C (Danchin and Sekowska, 2000). Third, natural competence appears in a batch culture of exponentially growing

cells; herein we mimic a growing population by letting the signal peptide concentration reflect the cell density in the culture.

In the initial model, only the core components of the system are taken into consideration, i.e., histidine kinase, histidine kinase in complex with CSP, response regulator and phosphorylated response regulator. The variables denote the total amount of the components in a population of n cells, and are described in Table 1, whereas the parameter definitions can be found in Table 2. The extracellular concentration of CSP is represented as a variable parameter (CSP) in order to study how the system responds to an increased signal peptide concentration in the medium.

Low amounts of mRNA transcripts of the *comCDE* operon have been detected in gene expression analysis, even though the quorum sensing system is not activated (Peterson et al., 2000). Therefore, ComD and ComE are assumed to be synthesized at a low basal rate described by the parameter β_i . Furthermore, phosphorylated ComE (ComE~P) acts as a transcription factor activating the *comCDE* promoter thus increasing the expression rate significantly (Morrison, 1997; Peterson et al., 2000). In the model, the effect of ComE~P on the synthesis of *comCDE* is described by a so called Monod saturation function, since the transcription rates are assumed to follow Michaelis–Menten kinetics (Dockery and Keener, 2001). The ComE~P induced synthesis of a component X_i has an upper limit determined by v_i , whereas k_i describes the required concentration of ComE~P for half-maximal biosynthesis. All components are also assumed to decay naturally at some rate δ_i . In addition, for intracellular components, the decay rates also account for the dilution effect due to cell division.

Non-complexed histidine kinases can form complexes with extracellular CSP molecules via the law of mass action at rate α . There are some evidences that ComD forms dimers when activated (Robinson et al., 2000; Stock et al., 2000). However, this is not taken into consideration in this model. The complexes consisting

Table 2
Definition of the parameters

Parameter	Definition	Unit	Reference
β_{ComC}	Basal synthesis rate for ComC, per 10^6 cells	nmol min^{-1}	Assigned
β_{ComD}	Basal synthesis rate for ComD, per 10^6 cells	nmol min^{-1}	Assigned
β_{ComE}	Basal synthesis rate for ComE, per 10^6 cells	nmol min^{-1}	Assigned
v_{maxComC}	Maximal synthesis rate of ComC, per 10^6 cells	nmol min^{-1}	Lee and Morrison (1999)
v_{maxComD}	Maximal synthesis rate of ComD, per 10^6 cells	nmol min^{-1}	Lee and Morrison (1999)
v_{maxComE}	Maximal synthesis rate of ComE, per 10^6 cells	nmol min^{-1}	Lee and Morrison (1999)
v_{maxComX}	Maximal synthesis rate of ComX, per 10^6 cells	nmol min^{-1}	Assigned
k_{ComC}	Required concentration of ComE~P for half-maximal synthesis rate of ComC	nmol min^{-1}	Assigned
k_{ComD}	Required concentration of ComE~P for half-maximal synthesis rate of ComD	nmol min^{-1}	Assigned
k_{ComE}	Required concentration of ComE~P for half-maximal synthesis rate of ComE	nmol min^{-1}	Assigned
k_{ComX}	Required concentration of ComE~P for half-maximal synthesis rate of ComX	nmol min^{-1}	Assigned
CSP	Extracellular concentration of CSP	pg ml^{-1}	Coomaraswamy (1996)
v_{int}	Volume of 10^6 cells	nl	Assigned
δ_{ComC}	Decay rate of ComC	min^{-1}	Assigned
δ_{ComD}	Decay rate of ComD	min^{-1}	Assigned
δ_{ComE}	Decay rate of ComE	min^{-1}	Ween et al. (1999)
δ_{CSP}	Decay rate of CSP	min^{-1}	Coffey et al. (1998)
δ_{ComDP}	Decay rate of ComDP	min^{-1}	Ween et al. (1999)
$\delta_{\text{ComE~P}}$	Decay rate of ComE~P	min^{-1}	Ween et al. (1999)
δ_{ComX}	Decay rate of ComX	min^{-1}	Lee and Morrison (1999)
δ_{R}	Decay rate of putative repressor	min^{-1}	Assigned
$v_{\text{max_export}}$	Maximal export rate of ComC, per 10^6 cells	nmol min^{-1}	Assigned
k_{export}	Required concentration of ComC for half-maximal export rate	nmol nl^{-1}	Assigned
α	Complex formation rate of CSP and ComD	$\text{ml min}^{-1} \text{pg}^{-1}$	Assigned
ψ	Scaling factor	pg nmol^{-1}	Assigned
γ	Dissociation rate of CSP and ComD	min^{-1}	Assigned
λ	Phosphorylation rate	$\text{nmol}^{-1} \text{min}^{-1}$	Assigned
ρ	Dephosphorylation rate	min^{-1}	Assigned
v_{ext}	Extracellular volume in the batch culture	ml	Assigned
n	Local cell density	$10^6 \text{ cells ml}^{-1}$	Assigned
v_{maxR}	Maximal synthesis rate of putative repressor, per 10^6 cells	nmol min^{-1}	Assigned
k_{R}	Required concentration of ComX for half-maximal synthesis rate of putative repressor	nmol nl^{-1}	Assigned
ω	Effect of putative repressor	–	Assigned
η	Hill coefficient	–	Assigned

of CSP molecules and histidine kinases dissociate at rate γ . The equation for the amount of ComD (X_{ComD}) can be expressed as:

$$\begin{aligned}
 \frac{dX_{\text{ComD}}}{dt} = & \underbrace{\beta_{\text{ComD}} n}_{\text{Basal synthesis}} \\
 & + \underbrace{v_{\text{maxComD}} n \left(\frac{X_{\text{ComE~P}}}{X_{\text{ComE~P}} + n v_{\text{int}} k_{\text{ComD}}} \right)}_{\text{ComE~P induced synthesis}} \\
 & - \underbrace{\delta_{\text{ComD}} X_{\text{ComD}}}_{\text{Decay}} - \underbrace{\alpha X_{\text{ComD}} \text{CSP}}_{\text{Complex formation}} \\
 & + \underbrace{\gamma X_{\text{ComDP}}}_{\text{Complex dissociation}}. \quad (1)
 \end{aligned}$$

The equation describing the amount of complexed ComD (X_{ComDP}) includes processes of complex formation and dissociation, and decay:

$$\begin{aligned}
 \frac{dX_{\text{ComDP}}}{dt} = & \underbrace{\alpha X_{\text{ComD}} \text{CSP}}_{\text{Complex formation}} - \underbrace{\gamma X_{\text{ComDP}}}_{\text{Complex dissociation}} \\
 & - \underbrace{\delta_{\text{ComDP}} X_{\text{ComDP}}}_{\text{Decay}}. \quad (2)
 \end{aligned}$$

The assumptions for synthesis and decay of ComE are the same as for ComD. The response regulator ComE is phosphorylated by complexed histidine kinase at rate λ , whereas dephosphorylation of ComE is assumed to occur spontaneously at rate ρ . The equations describing the changes in the total amounts of ComE (X_{ComE}) and

ComE~P ($X_{\text{ComE}\sim\text{P}}$) can thus be formalized as:

$$\frac{dX_{\text{ComE}}}{dt} = \underbrace{\beta_{\text{ComE}} n}_{\text{Basal synthesis}} + \underbrace{v_{\text{maxComE}} n \left(\frac{X_{\text{ComE}\sim\text{P}}}{X_{\text{ComE}\sim\text{P}} + n v_{\text{int}} k_{\text{ComE}}} \right)}_{\text{ComE}\sim\text{P induced synthesis}} - \underbrace{\delta_{\text{ComE}} X_{\text{ComE}}}_{\text{Decay}} - \underbrace{\frac{\lambda}{n v_{\text{int}}} X_{\text{ComE}} X_{\text{ComDP}}}_{\text{Phosphorylation of ComE}} + \underbrace{\rho X_{\text{ComE}\sim\text{P}}}_{\text{Dephosphorylation of ComE}\sim\text{P}}. \quad (3)$$

$$\frac{dX_{\text{ComE}\sim\text{P}}}{dt} = \underbrace{\frac{\lambda}{n v_{\text{int}}} X_{\text{ComDP}} X_{\text{ComE}}}_{\text{Phosphorylation of ComE}} - \underbrace{\rho X_{\text{ComE}\sim\text{P}}}_{\text{Dephosphorylation of ComE}\sim\text{P}} - \underbrace{\delta_{\text{ComE}\sim\text{P}} X_{\text{ComE}\sim\text{P}}}_{\text{Decay}}. \quad (4)$$

The calculations performed in this work are made with Matlab v. 6.5 (Mathworks Inc.) or Mathematica v. 3.0 (Wolfram Research Inc.).

2.2. Parameter estimations

The parameter values are assigned, as far as possible, from published data. In several cases the literature can only give a hint about a parameter value, for example, when estimating synthesis rates for the proteins only relative expressions of mRNA levels can be found. Although changes in RNA levels are measured precisely, the biological significance of an increase in mRNA remains an open question. As an example, in competent cells, *recA* mRNA has been estimated to increase up to 30-fold, whereas the RecA protein itself increases only fourfold. This increase causes, however, over 90% of the yield of recombinants (Peterson et al., 2000; Mortier-Barrière et al., 1998).

Since several quorum sensing models published earlier (Dockery and Keener, 2001; Anguige et al., 2004; Fagerlind et al., 2003; Gustafsson et al., 2004) have demonstrated hysteretic behaviors, it seems reasonable to assume that hysteretic dynamics also exist in the pneumococcal *com* system. Therefore, a parameter set that seems biologically plausible and that gives rise to hysteresis is chosen for the initial model. For those parameters where experimental data cannot be found, the parameter estimations are guided partly by the relation between parameter values used in earlier quorum sensing models (Dockery and Keener, 2001; Anguige et al., 2004; Fagerlind et al., 2003; Gustafsson et al., 2004), and partly by considering the plausibility of parameter values (Beckwitt et al., 1996). Furthermore, in the parameter estimation process, all parameters are tested in a wide range of values. However, the qualitative behavior of a system is determined chiefly by the network structure rather than by specific parameter values, i.e., the structure of the equations enables a certain qualitative

behavior, for example hysteresis and oscillations, but the parameter values determine whether the behaviors arise or not. Lastly, it should be emphasized that the dynamics described in the following sections only characterize the behaviors of our models with selected parameters, i.e., not all models and not experimental results.

2.3. Hysteretic bistability in quorum sensing regulated competence

The CSP concentration in the growth medium is assumed to be proportional to the cell density, so by increasing CSP we mimic a growing bacterial batch

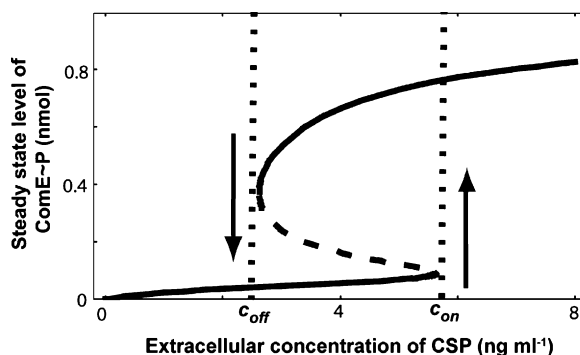


Fig. 2. Steady state amount of ComE~P ($X_{\text{ComE}\sim\text{P}}$) as a function of extracellular concentration of CSP. The parameter CSP provides a switch between the two stable steady states (solid line). At low concentrations of CSP, ComE~P has a low steady state level. As CSP has reached a certain threshold value (switch point c_{on} in the diagram), the steady state level of ComE~P switches to a higher value. The steady state amount of ComE~P will only go back to the lower steady state, if the concentration of CSP decreases enough and passes another switch point (switch point c_{off} in the diagram). The dashed line indicates unstable steady states. The parameter values used are $\beta_{\text{ComD}} = 0.025$, $\beta_{\text{ComE}} = 0.025$, $v_{\text{maxComD}} = 0.3$, $v_{\text{maxComE}} = 0.3$, $k_{\text{ComD}} = 0.7$, $k_{\text{ComE}} = 0.7$, $v_{\text{int}} = 1$, $\delta_{\text{ComD}} = 0.2$, $\delta_{\text{ComDP}} = 0.2$, $\delta_{\text{ComE}} = 0.035$, $\delta_{\text{ComE}\sim\text{P}} = 0.035$, $\alpha = 0.001$, $\gamma = 0.3$, $\lambda = 0.09$, $\rho = 0.4$, $v_{\text{ext}} = 1$, and $n = 1$.

culture. Increasing the extracellular signal peptide concentration affects the steady state level of phosphorylated response regulator in the cell (Fig. 2). At low concentrations of CSP, the steady state level of ComE~P is low. As CSP has reached a certain threshold concentration (indicated by c_{on} in Fig. 2), the level of ComE~P increases significantly, meaning that the *com* system becomes activated. The system only inactivates, turn-off, if the concentration of CSP decreases to levels significantly lower than the concentration required to activate the system (indicated by c_{off} in Fig. 2). Changing the parameter CSP, the so called control parameter, may switch the system state between the two stable branches. Such discontinuous toggle switches are termed hysteresis, which briefly also can be described as the lag between making a change and the response or effect of that change. Switch points, like these (c_{on} and c_{off} in Fig. 2), wherein a qualitative change in the behavior of non-linear system occurs when moving in the parameter space, are referred to as bifurcation points. More specifically, this type of bifurcation where a stable and an unstable steady state coalesce as in Fig. 2 is termed fold or saddle-node bifurcation.

2.4. Identification of a negative control mechanism

One aim with the study is to point out possible negative feedback mechanism(s) in the competence-evoking system in *S. pneumoniae*. It has been demonstrated experimentally, that the extracellular CSP concentration in a batch culture remains high a long time after the competence turn-off, and that the pneumococcal cells

are “blind” to CSP for approximately 1.5 generations (Morrison, 1997). From that, a decreasing CSP concentration can be excluded as shut-off mechanism. By performing an elasticity analysis, the impact of individual parameters on the system is examined and thereby providing an indication about possible negative regulators.

The elasticity analysis is therefore performed in the fold bifurcations of the hysteresis curve by computing the elasticity of each parameter, i.e., the relative change of the fold bifurcation value in relation to the relative change in the parameter value. Only the parameters representing biological processes (see Table 3) are taken into consideration in the elasticity analysis. In order to differentiate between control mechanisms occurring at transcriptional and translational levels at the biosynthesis, the elasticities of the synthesis parameters included in the term of ComE~P induced synthesis are investigated in two ways. First, if the regulation occurs at translational level, only the synthesis rate for that specific protein will be influenced. In the elasticity analysis, this is achieved by examining the elasticity for each synthesis parameter (i.e., v_i and k_i) separately. Second, if a control mechanism occurs at transcriptional level, for example a repressor acting on an operon, it will affect the synthesis for all gene products of that operon. The analysis is therefore performed with pooled synthesis parameters (i.e., v_{maxComD} and v_{maxComE} are assembled to one parameter, v_{maxComDE} , whereas k_{ComD} and k_{ComE} are assembled to k_{ComDE}) instead, making it possible to investigate the elasticities for control mechanisms occurring at transcriptional level.

Table 3
Elasticities in the fold bifurcations of the hysteresis curve for the initial model

Parameter	Elasticity in the off-switching fold (c_{off}) ^{a,b}	Elasticity in the on-switching fold (c_{on}) ^{a,b}
β_{ComD}	1.21	5.14
β_{ComE}	1.34	5.43
v_{maxComD}	5.03	7.12
v_{maxComE}	5.60	7.52
k_{ComD}	−3.29	−6.30
k_{ComE}	−3.66	−6.66
δ_{ComD}	−1.00	−1.00
δ_{ComDP}	−5.64	−11.7
δ_{ComE}	−6.24	−12.3
$\delta_{\text{ComE~P}}$	−1.20	−1.68
α	1.00	1.00
γ	−0.600	−0.600
λ	6.24	12.3
ρ	−5.74	−11.3
v_{maxComDE}	10.6	14.6
k_{ComDE}	−6.94	−13.0

^a The elasticities are presented with three significant digits.

^b Index (c_{off}) and (c_{on}) refer to the fold bifurcations marked in Fig. 2.

The elasticity analysis is performed in both fold bifurcations; however, the elasticity in the off-switching fold is considered most relevant as we are interested in pointing out a negative control mechanism. The absolute value for each parameter indicates its elasticity; the larger absolute value the more sensitive is the system in the fold bifurcations for changes occurring in that parameter. For a description of the technical details regarding the elasticity analysis, see [Appendix A](#). According to the elasticity analysis, the on/off-switching folds seem to be most sensitive to changes occurring at the transcriptional level ([Table 3](#)), since the elasticity is largest for the parameters $v_{\max\text{ComDE}}$ and k_{ComDE} . Other parameters that might be considered as potential shut-down mechanisms are, for example, the parameters λ and ρ , representing phosphorylation and dephosphorylation of response regulator, respectively. Interestingly, the mechanisms indicated to be most likely according to the result of the elasticity analysis are more or less in consistency with the hypotheses suggested from biological findings. Nonetheless, we decide to select one of several possible mechanisms for further evaluation. Guided by the result of the elasticity analysis besides biological plausibility, we suggest that the down regulation of the competence system possibly occurs at the transcriptional level by a putative repressor acting on the *comCDE* operon.

2.5. Feedback of CSP synthesis in the competence-evoking system

In the next step, we extend the model by including the synthesis of CSP to see how the system would behave in a population of increasing cell density. The synthesis and decay of the signal peptide, encoded by *comC*, is expressed mathematically in terms of ComD and ComE. The export of CSP is assumed to follow Michaelis–Menten kinetics. The upper rate limit for export is defined by the parameter $v_{\max_export}$, whereas the parameter k_{export} denotes the required amount of ComC for half-maximal export rate. The resulting model includes two additional equations describing the changes in the total amount of intracellular ComC (X_{ComC}) and in the concentration of extracellular CSP (X_{CSP}). The new equations added to the ODE system are:

$$\frac{dX_{\text{ComC}}}{dt} = \underbrace{\beta_{\text{ComC}} n}_{\text{Basal synthesis}} + \underbrace{v_{\max\text{ComC}} n \left(\frac{X_{\text{ComE~P}}}{X_{\text{ComE~P}} + n v_{\text{int}} k_{\text{ComC}}} \right)}_{\text{ComE~P induced synthesis}}$$

$$\begin{aligned} & \underbrace{-\delta_{\text{ComC}} X_{\text{ComC}}}_{\text{Decay}} \\ & \underbrace{-v_{\max_export} n \left(\frac{X_{\text{ComC}}}{X_{\text{ComC}} + n v_{\text{int}} k_{\text{export}}} \right)}_{\text{Export of precursor CSP}} \end{aligned} \quad (5)$$

$$\begin{aligned} \frac{dX_{\text{CSP}}}{dt} = & \underbrace{v_{\max_export} \frac{n}{v_{\text{ext}}} \psi \left(\frac{X_{\text{ComC}}}{X_{\text{ComC}} + n v_{\text{int}} k_{\text{export}}} \right)}_{\text{Export of precursor CSP}} \\ & \underbrace{-\alpha \psi X_{\text{CSP}} \frac{X_{\text{ComD}}}{v_{\text{ext}}}}_{\text{Complex formation}} + \underbrace{\gamma \psi \frac{X_{\text{ComDP}}}{v_{\text{ext}}}}_{\text{Complex dissociation}} \\ & \underbrace{-\delta_{\text{CSP}} X_{\text{CSP}}}_{\text{Decay}} \end{aligned} \quad (6)$$

Note that the parameter CSP has now been replaced with the variable X_{CSP} in the equations. In contrast to the initial model, we use the parameter n (population density) as a control parameter. In accordance with the hysteretic dynamics demonstrated in the initial model,

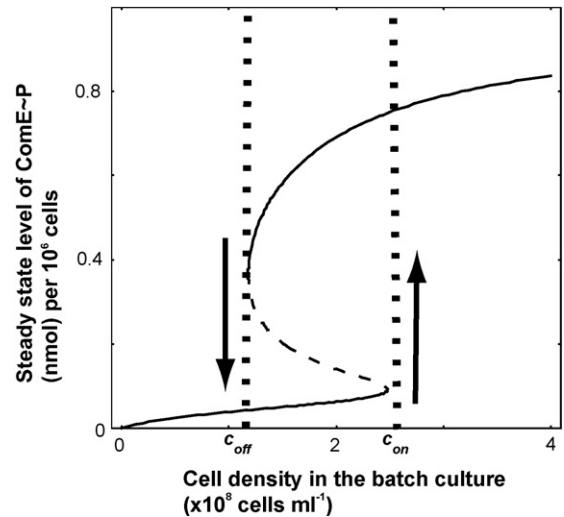


Fig. 3. The effect of varying the parameter n (cell density) on existence and stability property of a steady state amount of ComE~P ($X_{\text{ComE~P}}$) per 10^6 cells ml^{-1} . Here, the cell density provides the switch between the two stable steady states (solid line). When reaching a high enough cell density, the amount of ComE~P will jump up to a higher steady state (at a density around 2.5×10^8 cells ml^{-1} ; switch point c_{on} in the diagram). The off-switching fold is found at a density around 1×10^8 cells ml^{-1} (switch point c_{off} in the diagram). The dashed line indicates unstable steady states. Parameter settings are as in [Fig. 2](#) except additional parameters $\beta = 0.025$, $v_{\max\text{ComC}} = 0.3$, $k_{\text{ComC}} = 0.7$, $\delta_{\text{ComC}} = 0.2$, $\psi = 1$, $v_{\max_export} = 1$, and $k_{\text{export}} = 2$.

the extended model shows the same behavior (Fig. 3). The steady state level of phosphorylated response regulator depends on the population density in the batch culture. As the cell density increases, the level of ComE~P increases gradually until a critical value of the density is reached. At this fold (indicated by c_{on} in Fig. 3), the amount of ComE~P increases drastically and the system is activated. Once on the higher steady state branch, the system stays active even if the cell density falls. However, if the cell density decreases to values below the off-switching fold (indicated by c_{off} in Fig. 3), the system is turned off. This behavior is in agreement with the view that the quorum sensing regulation of natural competence is a way to monitor the concentration of potential gene donors in the immediate neighborhood.

2.6. Negative feedback in quorum sensing circuit

So far, we have only studied the initiation of competence. However, the competence is a transient state in *S. pneumoniae* so a negative feedback has to be assumed. In the light of suggested hypotheses and due to the elasticity analysis, we choose to include a hypothetical repressor as a shut-off mechanism in the model. Since shut-down is possibly caused by a ComX-dependent gene product the modified model comprises both ComX and a repressor, which inhibits the ComE~P-induced biosynthesis. In non-competent cells, ComX mRNA has not been detected (Luo and Morrison, 2003). Therefore, the basal synthesis of ComX in the model is approximated to zero. Expression of *comX* depends on ComE~P and is described mathematically by a Monod saturation function. The upper rate limit for ComX synthesis is defined by the parameter $v_{maxComX}$, whereas the parameter k_{ComX} denotes the required amount of ComE~P for

half-maximal synthesis rate. The decay of ComX occurs at rate δ_{ComX} . In summary, the equation for the amount of ComX (X_{ComX}) is expressed as:

$$\frac{dX_{ComX}}{dt} = \overbrace{v_{maxComX}n \left(\frac{X_{ComE\sim P}}{X_{ComE\sim P} + nv_{int}k_{ComX}} \right)}^{\text{ComE}\sim\text{P induced synthesis}} - \underbrace{\delta_{ComX}X_{ComX}}_{\text{Decay}}. \quad (7)$$

The expression of the putative repressor is assumed to depend on the alternative sigma factor ComX (Peterson et al., 2000). The synthesis of the repressor is described by a Hill function where the Hill coefficient, η , is interpreted biologically as the degree of positive cooperativity. The parameter v_{maxR} defines the upper limit for synthesis rate and the parameter k_R the required amount of ComX for half-maximum synthesis rate. The equation describing the total amount of repressor, X_R , can be summarized as:

$$\frac{dX_R}{dt} = \overbrace{v_{maxR}n \left(\frac{X_{ComX}^\eta}{X_{ComX}^\eta + nv_{int}k_R^\eta} \right)}^{\text{ComX dependent synthesis}} - \underbrace{\delta_R X_R}_{\text{Decay}}. \quad (8)$$

The repressive effect on the ComE~P-induced synthesis is accomplished by replacing the parameters k_{ComC} , k_{ComD} , k_{ComE} , and k_{ComX} in the equations by the expression ωX_R , where the parameter ω defines the effect of the repressor. Since the genes encoding CSP, ComD, and ComE are located on the same operon, and in lack of experimental data, the repressive effect is approximated to be equal for each individual component. The extended model comprises Eqs. (1)–(8), and the variables herein denote the total amount of the components

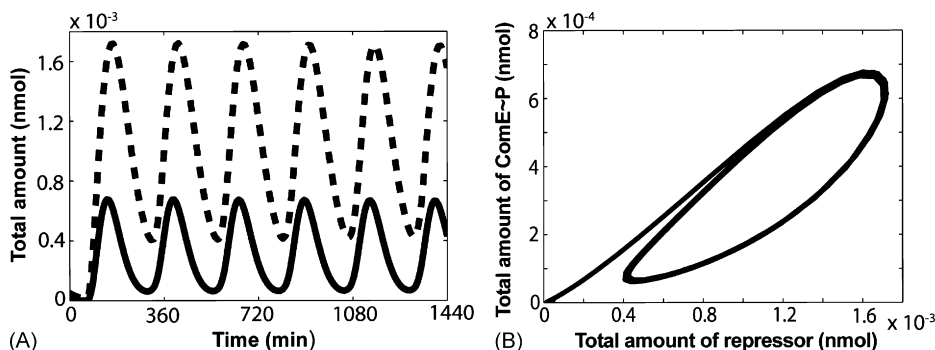


Fig. 4. Sustained oscillations in the competence system. In (A), a numerical solution for the system over time is shown. The dashed line indicates the total amount of repressor, whereas the solid line indicates the total amount of ComE~P. In (B), a phase diagram for the total amounts of phosphorylated response regulator ($X_{ComE\sim P}$) and repressor (X_R) is presented. The system tends to a stable limit cycle. Parameter values used are as in Fig. 3 except $\beta_{ComC} = 0.006$, $\beta_{ComD} = 0.006$, $\beta_{ComE} = 0.006$, $v_{maxComX} = 0.2$, $k_{ComX} = 0.7$, $\delta_{ComX} = 0.2$, $v_{maxR} = 0.0003$, $k_R = 0.6$, $\delta_R = 0.023$, and $\eta = 2.4$.

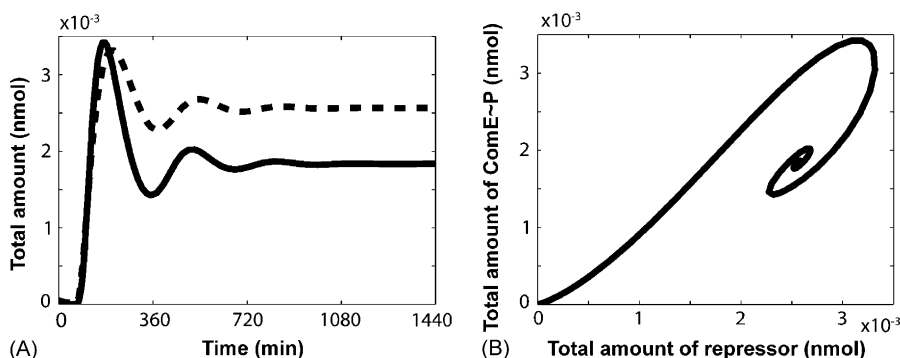


Fig. 5. Damped oscillations in the competence system. A numerical solution for the system over time is visualized in (A) where the dashed line indicates the total amount of repressor, and the solid line indicates the total amount of ComE~P. In (B), the corresponding phase diagram is shown. The system passes a supercritical Hopf bifurcation when decreasing the value of the parameter $v_{\max R}$ from 0.0003 to 0.0002, and the dynamic in the system changes from sustained to damped oscillations. The unstable steady state, which the system oscillates around on a stable limit cycle at sustained oscillatory dynamic, changes its stability property when passing the bifurcation. The system tends to settle at this steady state. For parameter settings see Fig. 4, except for $v_{\max R} = 0.0002$.

in a population of constant cell density, as in the initial model.

Since oscillating dynamics in the pneumococcal competence system have been observed in experimental studies, we are interested to see if oscillations may appear when including a putative repressor as a negative control mechanism. Depending on parameter values, the system may exhibit damped or sustained oscillations. As shown in Fig. 4A, the amounts of ComE~P and repressor oscillate over time. In the projected phase diagram in Fig. 4B, a trajectory of the system is shown. The trajectory moves along a stable limit cycle around an unstable steady state. As the parameter $v_{\max R}$, i.e., the maximal synthesis rate for the repressor, is decreased (Fig. 5A), damped oscillations are envisaged by the model. This parameter adjustment makes the unstable steady state in the center of the closed orbit stable, so the system instead tends to settle on this steady state (Fig. 5B).

3. Discussion

The competence-evoking quorum sensing system in *S. pneumoniae* has been subject to investigations for a long time. Although extensive laboratory efforts, the mechanism(s) behind the abrupt competence shut-off remains unknown. Due to experimental difficulties to identify the negative regulation, we found it suitable to investigate the regulation of the pneumococcal *com* system by a mathematical approach.

Several mechanisms have been hypothesized as responsible for the competence shut-off. However, it has been shown that the complete competence shut-off in wild-type cells depends largely on the alternative sigma factor ComX (Peterson et al., 2000), so we

considered a hypothesis involving a ComX-dependent gene product as most attractive for the moment. Possibly, pneumococci have evolved a negative control mechanism where it would be most effective, i.e., least “cost” for the bacterium. So, by performing an elasticity analysis, we could examine the impact of individual parameters on the system’s hysteretic behavior, serving as guidance concerning possible negative regulators of the competence system. As expected, due to the linearity effect caused by using pooled synthesis parameters, the *com* system seemed to respond strongest to changes that occurred in processes at transcriptional level according to the analysis (Table 3). However, controlling transcription of a group of genes involved in the same process instead of regulating single genes is a more efficient strategy and is found commonly in prokaryotes. Based on the elasticity analysis, we suggest that the negative control in the competence system is possibly exerted by a repressor.

We introduced a hypothetical *comCDE* repressor in the model to see how the system would respond to this negative mechanism. According to experimental findings, we assumed that the putative repressor is a ComX-dependent gene product, which inhibits expression of *comCDE* and *comX*. The model was therefore extended to include both the alternative sigma factor ComX and the putative repressor. As it has been demonstrated in experimental studies that the *com* system exhibits some kind of oscillatory behavior (Chen and Morrison, 1998; Morrison, 1997), we were especially interested to examine the existence of oscillations in the competence system. Our model system predicted that the *com* system may exhibit sustained (Fig. 4) as well as damped (Fig. 5) oscillations depending on the maximal synthesis rate

for the repressor. Experimentally, the existence of more than one competence wave has been demonstrated being dependent on the initial pH in the medium (Chen and Morrison, 1998). In cultures grown in media of lower initial pH, competence occurred only once and at higher cell densities, whereas in more alkaline media a succession of competence waves occurred, beginning at lower cell densities. Possibly, the culture's entrance into stationary phase sets bound the number of competence waves. Besides, the appearance of damped oscillations in batch cultures (Chen and Morrison, 1998; Morrison, 1997) may also be due to more and more cells entering stationary phase. Since different growth phases were not considered in the model, the achieved *in silico* results can be seen as the dynamic of the competence system when not being constrained by other factors. Additionally, experimental competence is often measured as DNA digestion activity as done by Chen and Morrison (1998). Since our competence model only consider total protein amounts in a constant population, it is difficult to directly compare our *in silico* results with results of experimental studies of the *com* system. For testing the existence of a repressor acting on the *comCDE* and *comX* operons, we propose an experimental intervention where regulatory proteins, that bind to promoter fragments of the operons coupled to magnetic beads, are identified using matrix-assisted laser desorption ionization-time of flight (MALDI-TOF) (Tegmark et al., 2000).

Initially, we constructed a simple model only considering the core components of the system. According to many other biological systems, the pneumococcal competence-evoking system exhibited a switch-like behavior where a graded signal is turned into an all-or-nothing response. Such systems are often referred to as 'bistable'. In the model, the bistability corresponds to non-activated respective activated states of the *com* system. Some bistable circuits exhibit a hysteretic behavior, which gives a memory effect. This history dependence implies that the order of preceding events influence on the order of subsequent events. The state of the *com* system depends on the preceding state as shown in Fig. 2. Even if the control parameter decreases below the on-value, the system remains on the upper branch. The control parameter must decrease below the value indicated by ' c_{on} ' in Fig. 2. This memory effect occurs in the interval $c_{on}-c_{off}$ where the system is bistable. Although hysteretic behaviors have been demonstrated theoretically in several bacterial quorum sensing systems earlier (Dockery and Keener, 2001; Anguige et al., 2004; Fagerlind et al., 2003; Gustafsson et al., 2004), it has not been verified experimentally yet. This is probably due to the difficulties associated with such an experiment, e.g.

it requires a reporter gene product having an extremely short half-life.

Thereafter, the initial model was extended to include CSP synthesis in order to study the dynamic of the competence system in a growing bacterial population. Population density was utilized as a control parameter. As shown in Fig. 3, the extended system exhibited the same hysteretic dynamic as the initial model (Fig. 2). The *com* system was activated when the cell density in the batch culture reached a certain threshold, around 2.5×10^8 cells ml⁻¹. This threshold value corresponds well with experimental data (Chen and Morrison, 1998). Besides cell density, different aspects of environment and metabolism, such as temperature, pH (Chen and Morrison, 1998), cations (Fox and Hotchkiss, 1957), and oxygen concentration (Echenique et al., 2000) affect competence initiation. These additional factors explain the wide range in cell density at which competence can be initiated in a batch culture.

Nevertheless, CSP-independent, and thus cell density independent, competence has been described in streptococci. Ween et al. (2002) reported that a natural *Streptococcus infantis* isolate was competent even though it has lost its gene encoding CSP. Plausibly, as the authors proposed, mutations in *comD* or *comE* have resulted in a gain-of-function phenotype allowing competence without CSP. However, so far, such a CSP-independent competence induction has not been observed in any natural isolate of *S. pneumoniae*.

Possibly, a large number of negative regulators at multiple levels control the competence system in pneumococci as in other bacterial quorum sensing systems (Shuster and Greenberg, 2006; Ward et al., 2004). For example, competence development was shown to be inhibited by increased basal-level expression of *comDE* by Guiral et al. (2006). Furthermore, the protease ClpP appears to regulate competence negatively (Sung and Morrison, 2005) by degrading ComX. However, at present, neither of these mechanisms seems to account for the rapid ComX-dependence down-regulation of the early *com* genes as demonstrated by Peterson et al. (2000). Taken together, still many pieces remain to be fit together in the complex puzzle of the quorum-sensing network regulating competence in *S. pneumoniae*.

In this study, we have examined the regulation of the pneumococcal competence-evoking quorum sensing system. In particular, we focused on pointing out a possible molecular mechanism responsible for the abrupt shut-off of the *com* system. The results presented in this paper indicate that the pneumococcal competence system is possibly down-regulated at transcriptional level by a putative repressor inhibiting expression of *comCDE*.

and *comX*. Besides, we demonstrated that including such a repressor in the system may cause waves of competence in a batch culture, which is in consistency with experimental observations. A specific experimental intervention was also suggested in order to test the model prediction of the existence of a hypothetical repressor.

Acknowledgments

We thank Andreas Jansson for helpful discussions, and Mikael Ejdebäck for critical reading of the manuscript. This work has been supported by Skaraborg Hospital and by the Swedish Knowledge Foundation through the Industrial PhD program in Medical Bioinformatics at the Strategy and Development Office (SDO) at Karolinska Institutet.

Appendix A. Description of technical issues in computing the elasticity of the switch points

The steady state equations of the system are given by equating to zero all derivatives in the differential equation (DE) system. The right-hand sides of the DEs are rational functions in the variables and parameters. Multiplying with common denominators yields a system of algebraic equations, whose solutions are the steady states of the DE system. For each parameter set, the solution is multivalued, and the number of real solutions depends on the parameter values. We are interested in finding the points in the parameter space, where the system may switch between active and inactive state. Mathematically these points are fold (or saddle-node) bifurcations of the steady state system, where moving in the parameter space makes a stable and an unstable steady state coalesce and disappear.

We choose a reporter component of the system; in our case *ComE~P*, to define the activity of the system. The Buchberger algorithm, implemented by the function *GroebnerBasis* in Mathematica (Wolfram Research Inc.), makes it possible to eliminate the other components of the system, yielding a single equation in the amount of reporter component, *x*, a control parameter *c* (in our case concentration of CSP), and all other model parameters $p_1, \dots, p_n$:

$$f(x, c, p_1, \dots, p_n) = 0. \quad (\text{A.1})$$

For fixed values of $p_1, \dots, p_n$ this yields a response curve in the *cx*-plane as in Fig. 2. Folds appear where the curve has a vertical tangent, which is the case when:

$$\frac{\partial f}{\partial x}(x, c, p_1, \dots, p_n) = 0. \quad (\text{A.2})$$

Eliminating *x* from these two equations gives an equation:

$$g(c, p_1, \dots, p_n) = 0 \quad (\text{A.3})$$

whose solutions are the points in parameter space, where there are fold bifurcations or other bifurcations in the system. We may solve Eq. (A.3) for *c*. The solution is generally multi-valued. For values of $p_1, \dots, p_n$ where the *c* response shows hysteresis, i.e., has an S-shaped response curve, there are two corresponding solutions. This defines locally two functions, $c_{\text{on}}(p_1, \dots, p_n)$ and $c_{\text{off}}(p_1, \dots, p_n)$, the values of *c* where the system switches from off to on and conversely, for given values of the independent parameters $p_1, \dots, p_n$.

We will now concentrate on the elasticity of c_{on} and c_{off} to changes in the independent parameters. The elasticity of c_{on} with respect to the independent parameter p_k is:

$$\varepsilon_{c_{\text{on}}, p_k} = \frac{\partial(\log c_{\text{on}})}{\partial(\log p_k)} = \frac{p_k}{c_{\text{on}}} \frac{\partial c_{\text{on}}}{\partial p_k} \quad (\text{A.4})$$

and accordingly for c_{off} .

Implicit differentiation in the equation $g(c, p_1, \dots, p_n) = 0$ with respect to each independent parameter gives linear equations for the elasticities, and we can solve for the each of them. The value of the elasticity ε_{c, p_k} is the relative change of the switch point value of *c* in relation to the relative change in value of p_k , while other parameters are fixed. Note that this function depends on parameter values. The elasticities will thus help us in pointing out the mechanisms in the system that most effectively alters the systems response to CSP.

References

- Anguige, K., King, J.R., Ward, J.P., Williams, P., 2004. Mathematical modeling of therapies targeted at bacterial quorum sensing. *Math. Biosci.* 192, 39–83.
- Bartilson, M., Marra, A., Christine, J., Asundi, J.S., Schneider, W.P., Hromockyj, A.E., 2001. Differential fluorescence induction reveals *Streptococcus pneumoniae* loci regulated by competence stimulatory peptide. *Mol. Microbiol.* 39, 126–135.
- Beckwith, J., Gralla, J.D., Collado-Vides, J., Ninfa, A.J., Landick, R., Turnbough Jr., C.L., Yanofsky, C., Choy, H., Adhya, S., Schlieff, R., Neidhardt, F.C., Savageau, M.A., Saier Jr., M.H., Ramseier, T.M., Reizer, J., Magasanik, B., Wanner, B.L., Gross, C.A., Walker, G.C., Keener, J., Nomura, M., Grunberg-Manago, M., Cashel, M., Gentry, D.R., Hernandez, V.J., Vinella, D., Hengge-Aronis, R., Newman, E.B., Lin, R.T., D'Ari, R., Lynch, A.S., Lin, E.C.C., Slonczewski, J.L., Foster, J.W., 1996. Regulation of gene expression. In: Neidhardt, F.C. (Ed.), *Escherichia coli* and *Salmonella*. Cellular and Molecular Biology, 2nd ed. ASM Press, Washington, DC, pp. 1225–1550.

- Campbell, E.A., Choi, S.Y., Masure, H.R., 1998. A competence regulon in *Streptococcus pneumoniae* revealed by genomic analysis. *Mol. Microbiol.* 27, 929–939.
- Chen, J.D., Morrison, D.A., 1998. Modulation of competence for genetic transformation in *Streptococcus pneumoniae*. *J. Gen. Microbiol.* 133, 1959–1967.
- Coffey, T.J., Dowson, C.G., Daniels, M., Zhou, J., Martin, C., Spratt, B.G., Musser, J.M., 1991. Horizontal transfer of multiple penicillin-binding protein genes, and capsule biosynthetic genes, in natural populations of *Streptococcus pneumoniae*. *Mol. Microbiol.* 5, 2255–2260.
- Coffey, T.J., Enright, M.C., Daniels, M., Morona, J.K., Morona, R., Hryniewicz, W., Paton, J.-C., Spratt, B.G., 1998. Recombinational exchanges at the capsular polysaccharide biosynthetic locus lead to frequent serotype changes among natural isolates of *Streptococcus pneumoniae*. *Mol. Microbiol.* 27, 73–83.
- Coomaraswamy, G., 1996. Induction of competence for genetic transformation in *Streptococcus pneumoniae* by a pheromone peptide and its synthetic analogues. PhD Thesis, University of Illinois at Chicago, Chicago.
- Danchin, A., Sekowska, A., 2000. Expression profiling in reference bacteria: dreams and reality. *Genome Biol.* 1, 1024.1–1024.5.
- Dockery, J.D., Keener, J.P., 2001. A mathematical model for quorum sensing in *Pseudomonas aeruginosa*. *Bull. Math. Biol.* 63, 95–116.
- Dowson, C.G., Hutchinson, A., Brannigan, J.A., George, R.C., Hansman, D., Linares, J., Thomasz, A., Maynard, J.M., Spratt, B.G., 1989. Horizontal transfer of penicillin-binding protein genes in penicillin-resistant clinical isolates of *Streptococcus pneumoniae*. *Proc. Natl. Acad. Sci. U.S.A.* 86, 8842–8846.
- Dunny, G.M., Leonard, B.A.B., 1997. Cell–cell communication in gram-positive bacteria. *Annu. Rev. Microbiol.* 51, 527–564.
- Echenique, J.R., Chapuy-Regaud, S., Trombe, M.-C., 2000. Competence regulation by oxygen in *Streptococcus pneumoniae*: involvement of *ciaRH* and *comCDE*. *Mol. Microbiol.* 36, 688–696.
- Fagerlind, M.G., Rice, S.A., Nilsson, P., Harlén, M., James, S., Charlton, T., Kjelleberg, S., 2003. The role of regulators in the expression of quorum-sensing signals in *Pseudomonas aeruginosa*. *J. Mol. Microbiol. Biotechnol.* 6, 88–100.
- Fox, M.S., Hotchkiss, R.D., 1957. Initiation of bacterial transformation. *Nature* 179, 1322–1325.
- Guiral, S., Mitchell, T.J., Martin, B., Claverys, J.-P., 2005. Competence-programmed predation of non-competent cells in the human pathogen *Streptococcus pneumoniae*: genetic requirements. *Proc. Natl. Acad. Sci. U.S.A.* 102, 8710–8715.
- Guiral, S., Hénard, V., Grandadel, C., Martin, B., Claverys, J.-P., 2006. Inhibition of competence development in *Streptococcus pneumoniae* by increased basal-level expression of the ComDE two-component regulatory system. *Microbiology* 152, 323–331.
- Gustafsson, E., Nilsson, P., Karlsson, S., Arvidsson, S., 2004. Characterizing the dynamics of the quorum-sensing system in *Staphylococcus aureus*. *J. Mol. Microbiol. Biotechnol.* 8, 232–242.
- Hakenbeck, R., König, A., Kern, I., van der Linden, M., Keck, W., Billot-Klein, D., Legrand, R., Schoot, B., Gutmann, L., 1998. Acquisition of five high- M_r penicillin-binding protein variants during transfer of high-level β -lactam resistance from *Streptococcus mitis* to *Streptococcus pneumoniae*. *J. Bacteriol.* 180, 1831–1840.
- Hollingshead, S.K., Becker, R., Briles, D.E., 2000. Diversity of PspA: mosaic genes and evidence for past recombination in *Streptococcus pneumoniae*. *Infect. Immun.* 68, 5889–5900.
- Hui, F., Zhou, M.L., Morrison, D.A., 1995. Competence for genetic transformation in *Streptococcus pneumoniae*: organization of a regulatory locus with homology to two lactococcal A secretion genes. *Gene* 153, 25–31.
- Håvarstein, L.S., Coomaraswamy, G., Morrison, D.A., 1995. An unmodified heptadecapeptide pheromone induces competence for genetic transformation in *Streptococcus pneumoniae*. *Proc. Natl. Acad. Sci. U.S.A.* 92, 11140–11144.
- Kelly, T., Dillard, J.P., Yother, J., 1994. Effect of genetic switching of capsular type on virulence of *Streptococcus pneumoniae*. *Infect. Immun.* 62, 1813–1819.
- Lau, G.W., Haataja, S., Lonetto, M., Kensit, S.E., Marra, A., Bryant, A.P., McDewitt, D., Morrison, D.A., Holden, D.W., 2001. A functional genomic analysis of type 3 *Streptococcus pneumoniae* virulence. *Mol. Microbiol.* 40, 555–571.
- Lee, M., Morrison, D.A., 1999. Identification of a new regulator in *Streptococcus pneumoniae* linking quorum sensing to competence for genetic transformation. *J. Bacteriol.* 181, 349–358.
- Luo, P., Morrison, D.A., 2003. Transient association of an alternative sigma factor, ComX, with RNA polymerase during the period of competence for genetic transformation in *Streptococcus pneumoniae*. *J. Bacteriol.* 185, 349–358.
- Luo, P., Li, H., Morrison, D.A., 2003. ComX is a unique link between multiple quorum sensing outputs and competence in *Streptococcus pneumoniae*. *Mol. Microbiol.* 50, 623–633.
- Martin, B., Garcia, P., Castanié, M.P., Claverys, J.P., 1995. The *recA* gene of *Streptococcus pneumoniae* is part of a competence-induced operon and controls lysogenic induction. *Mol. Microbiol.* 15, 367–379.
- Morrison, D.A., 1997. Streptococcal competence for genetic transformation: regulation by peptide pheromones. *Microb. Drug Res.* 3, 27–37.
- Mortier-Barrière, I., de Saizieu, A., Claverys, J.-P., Martin, B., 1998. Competence-specific induction of *recA* is required for full recombination proficiency during transformation in *Streptococcus pneumoniae*. *Mol. Microbiol.* 27, 159–170.
- Moscato, M., Claverys, J.-P., 2004. Release of DNA into the medium by competent *Streptococcus pneumoniae*: kinetics, mechanism and stability of the liberated DNA. *Mol. Microbiol.* 54, 783–794.
- Pestova, E.V., Håvarstein, L.S., Morrison, D.A., 1996. Regulation of competence for genetic transformation in *Streptococcus pneumoniae* by an auto-induced peptide pheromone and a two-component regulatory system. *Mol. Microbiol.* 21, 853–864.
- Pestova, E.V., Morrison, D.A., 1998. Isolation and characterization of three *Streptococcus pneumoniae* transformation-specific loci by use of a *lacZ* reporter insertion vector. *J. Bacteriol.* 180, 2701–2710.
- Peterson, S., Cline, R.T., Tettelin, H., Sharov, V., Morrison, D.A., 2000. Gene expression analysis of the *Streptococcus pneumoniae* competence regulons by use of DNA microarrays. *J. Bacteriol.* 182, 6192–6202.
- Peterson, S.N., Sung, C.K., Cline, R., Desai, B., Snesrud, E.C., Luo, P., Walling, J., Li, H., Mintz, M., Tsegaye, G., Burr, P.C., Do, Y., Ahn, S., Gilbert, J., Fleischmann, R.D., Morrison, D.A., 2004. Identification of competence pheromone responsive genes in *Streptococcus pneumoniae* by use of DNA microarrays. *Mol. Microbiol.* 51, 1051–1070.
- Robinson, V.L., Buckler, D.R., Stock, A.M., 2000. A tale of two components: a novel kinase and a regulatory switch. *Nat. Struct. Biol.* 7, 626–633.
- Shuster, M., Greenberg, E.P., 2006. A network of networks: quorum-sensing gene regulation in *Pseudomonas aeruginosa*. *Int. J. Med. Microbiol.* 296, 73–81.

- Stock, A.M., Robinson, V.L., Goudreau, P.N., 2000. Two-component signal transduction. *Annu. Rev. Biochem.* 69, 183–215.
- Sung, C.K., Morrison, D.A., 2005. Two distinct functions of ComW in stabilization and activation of the alternative sigma factor ComX in *Streptococcus pneumoniae*. *J. Bacteriol.* 187, 3052–3061.
- Tegmark, K., Karlsson, A., Arvidson, S., 2000. Identification and characterization of SarH1, a new global regulator of virulence gene expression in *Staphylococcus aureus*. *Mol. Microbiol.* 37, 398–409.
- Ward, J.P., King, J.R., Koerber, A.J., Croft, J.M., Sockett, R.E., Williams, P., 2004. Cell-signalling repression in bacterial quorum sensing. *Math. Med. Biol.* 21, 169–204.
- Ween, O., Gaustad, P., Håvarstein, L.S., 1999. Identification of DNA binding sites for ComE, a key regulator of natural competence in *Streptococcus pneumoniae*. *Mol. Microbiol.* 33, 817–827.
- Ween, O., Teigen, S., Gaustad, P., Kilian, M., Håvarstein, L.S., 2002. Competence without a competence pheromone in a natural isolate of *Streptococcus infantis*. *J. Bacteriol.* 184, 3426–3432.